SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0512

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)

Assessment Tool Weightage: 20%

1. Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from real-world scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner.
2. Develop a hierarchically structured concept map that identifies major stages of visual data handling in a Computer Vision system, clearly illustrates logical relationships and contributions of each stage toward generating meaningful outputs, incorporates relevant literature connections, and ensures clarity and readability.
3. Create a detailed concept map that accurately identifies concepts related to image resolution and intensity levels, establishes clear cause–effect relationships with image quality and usability, integrates supporting evidence from literature, and presents the concepts in a logically organized and visually clear format.
4. Design an analytically organized concept map that identifies key factors in image acquisition conditions, clearly links them to downstream processing and application performance through meaningful relationships, integrates findings from relevant studies, and ensures a well-structured and easy-to-interpret presentation.
5. Construct an original and logically structured concept map that identifies system capabilities and their influence on the selection and effectiveness of vision-based applications, establishes clear interconnections, integrates multiple literature perspectives, and maintains high clarity and visual organization.
6. Construct a concept map that identifies the complete workflow of digital image formation, clearly linking sensing, sampling, and quantization stages with their impact on image quality, integrating relevant literature insights, and presenting the relationships in a structured and visually coherent manner.
7. Develop a concept map that identifies different levels of Computer Vision, establishes relationships between low-level processing, mid-level analysis, and high-level interpretation, integrates literature-based examples, and ensures clear hierarchical organization.
8. Create a concept map that identifies factors affecting spatial resolution, clearly linking them to sampling rate, pixel density, and application requirements,

incorporating supporting evidence from literature, and presenting the structure with clarity.

9. Design a concept map that identifies the role of image sensing mechanisms, clearly establishing relationships between sensor characteristics, noise generation, and acquisition quality, integrating literature perspectives, and ensuring logical flow.
10. Construct a concept map that identifies the relationship between sampling, aliasing, and image distortion, clearly illustrating cause–effect dependencies, integrating relevant theoretical insights, and presenting the structure in a visually organized manner.
11. Develop a concept map that identifies quantization processes and their impact on intensity resolution, clearly linking quantization levels to image quality degradation, incorporating literature references, and ensuring structured visualization.
12. Create a concept map that identifies the relationship between image acquisition conditions (lighting, motion, environment) and resulting image artifacts, integrating findings from literature and presenting clear interconnections.
13. Design a concept map that identifies the transformation from analog scene to digital image, clearly linking physical processes to computational representation, integrating literature-based insights, and ensuring clarity and organization.
14. Construct a concept map that identifies dependencies between image quality parameters (resolution, noise, contrast) and system performance, integrating multiple perspectives from literature and presenting structured relationships.
15. Develop a concept map that identifies interactions between image formation models and Computer Vision applications, clearly linking theoretical models to practical system performance, integrating literature insights, and ensuring visual clarity.

Code of Conduct:

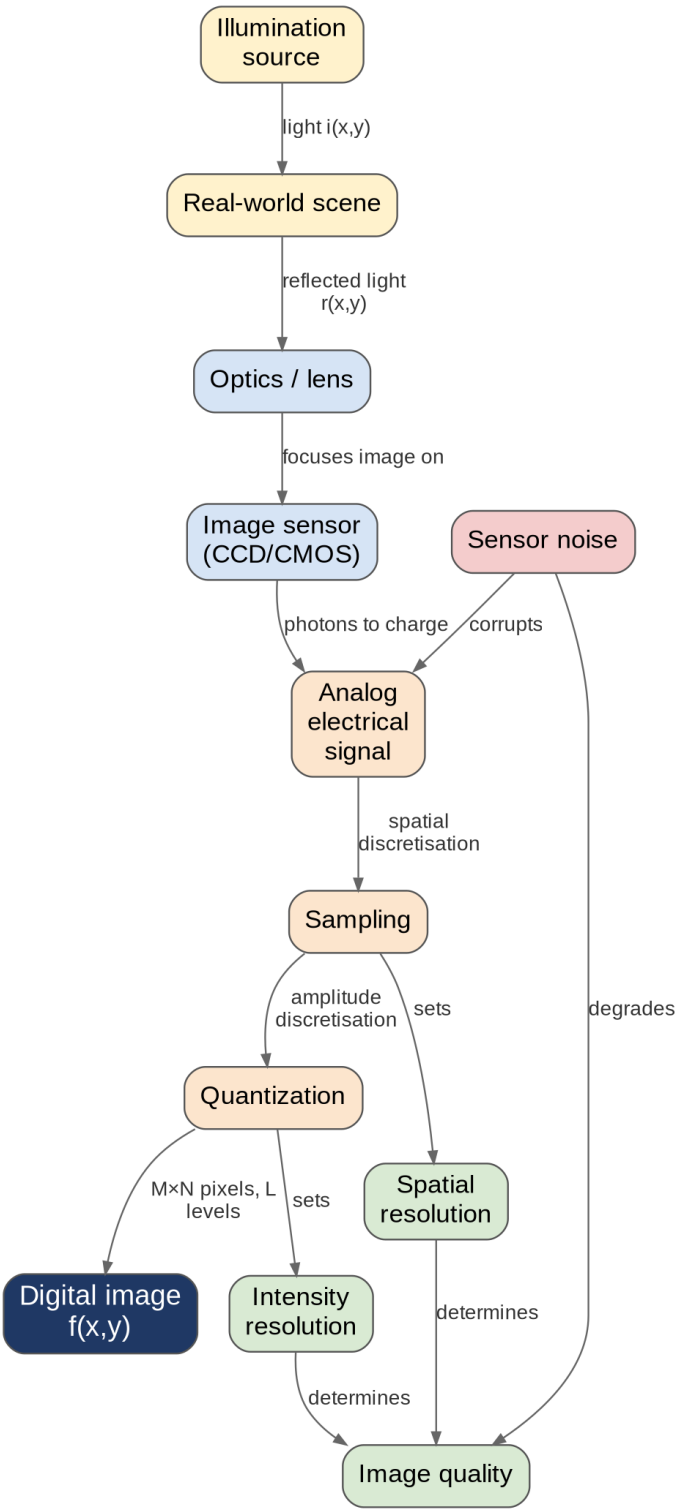
I **A.N.L.NARASIMHA(192525267)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

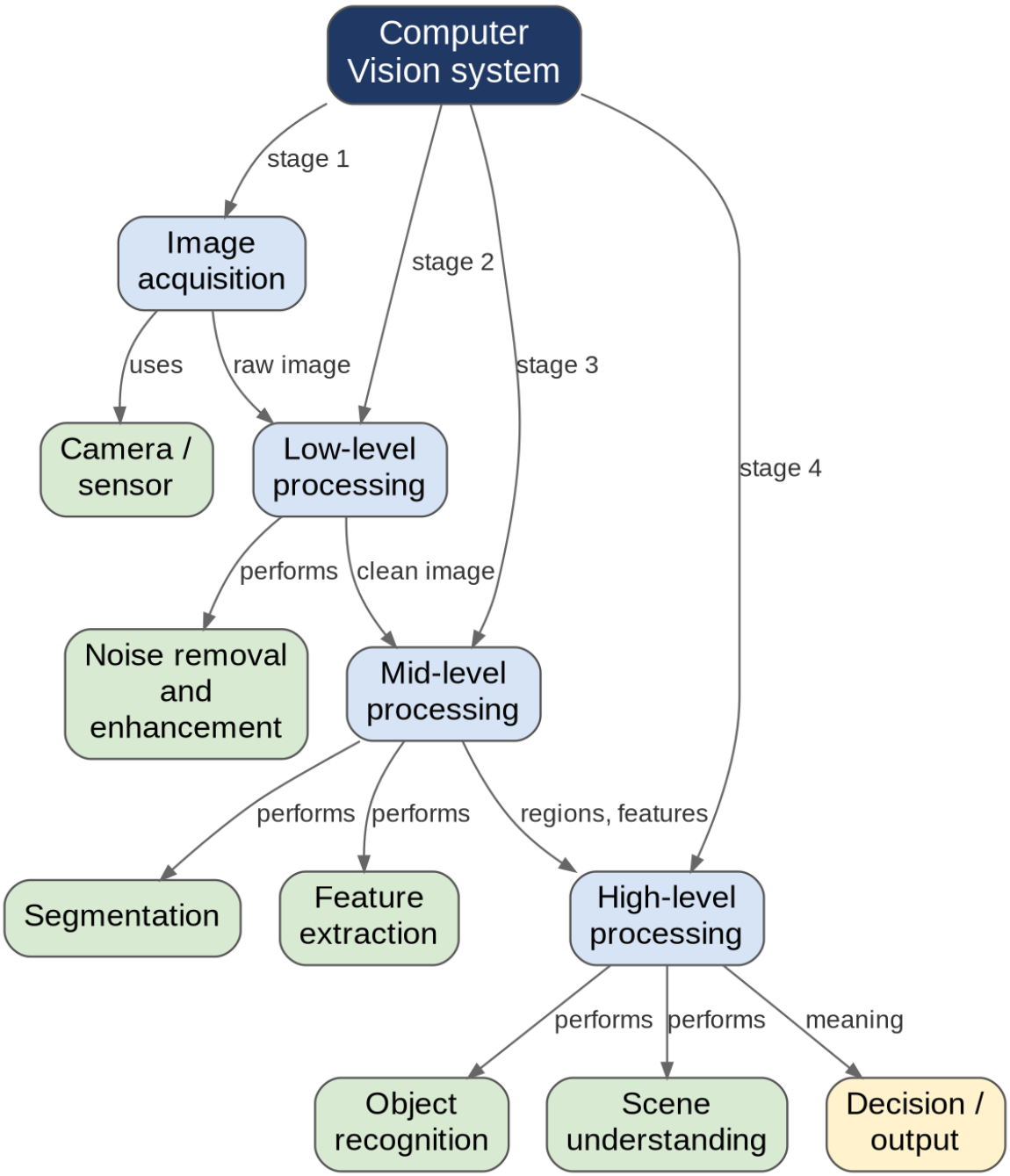
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Real-world scene to digital image pipeline



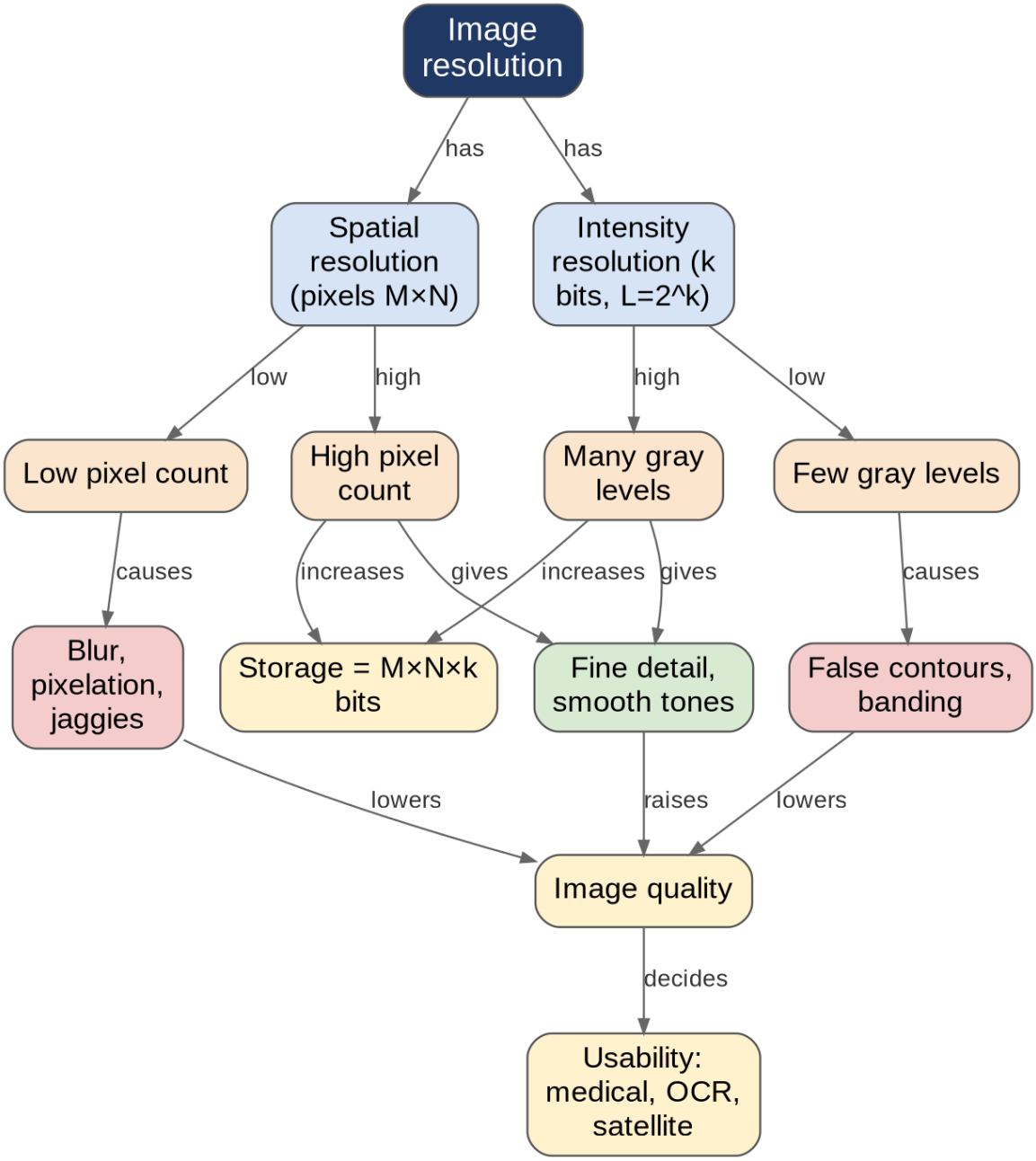
Literature link: *[1] presents the digital image formation chain (illumination, sensing, sampling, quantization); [2] covers the photometric and geometric image formation pipeline; [6] compares sensor read-out and noise.*

. Hierarchical stages of visual data handling



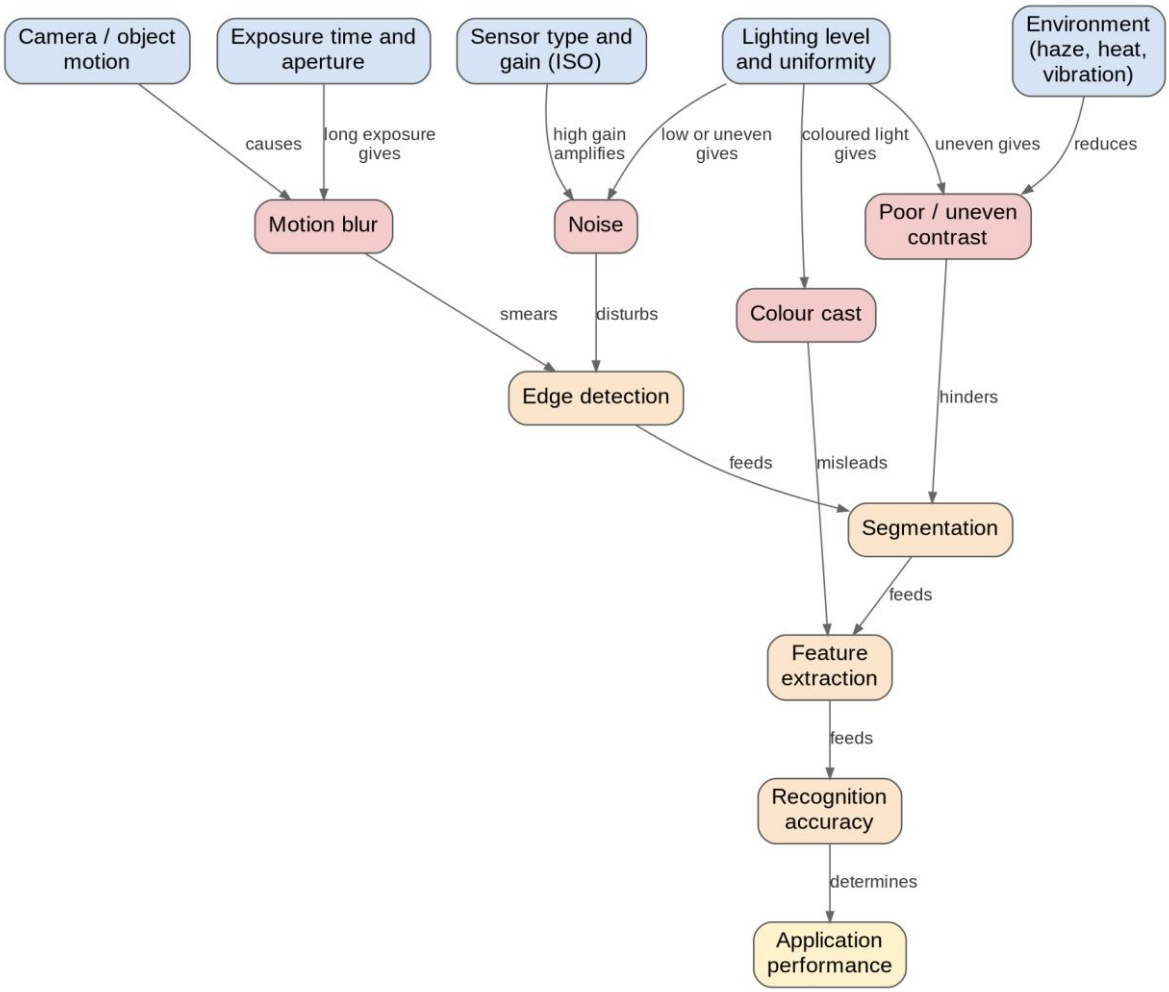
Literature link: *[3] and [2] organise vision into acquisition, processing, grouping and recognition stages; [4] gives the early-to-high-level hierarchy of visual processing.*

3. Image resolution and intensity levels



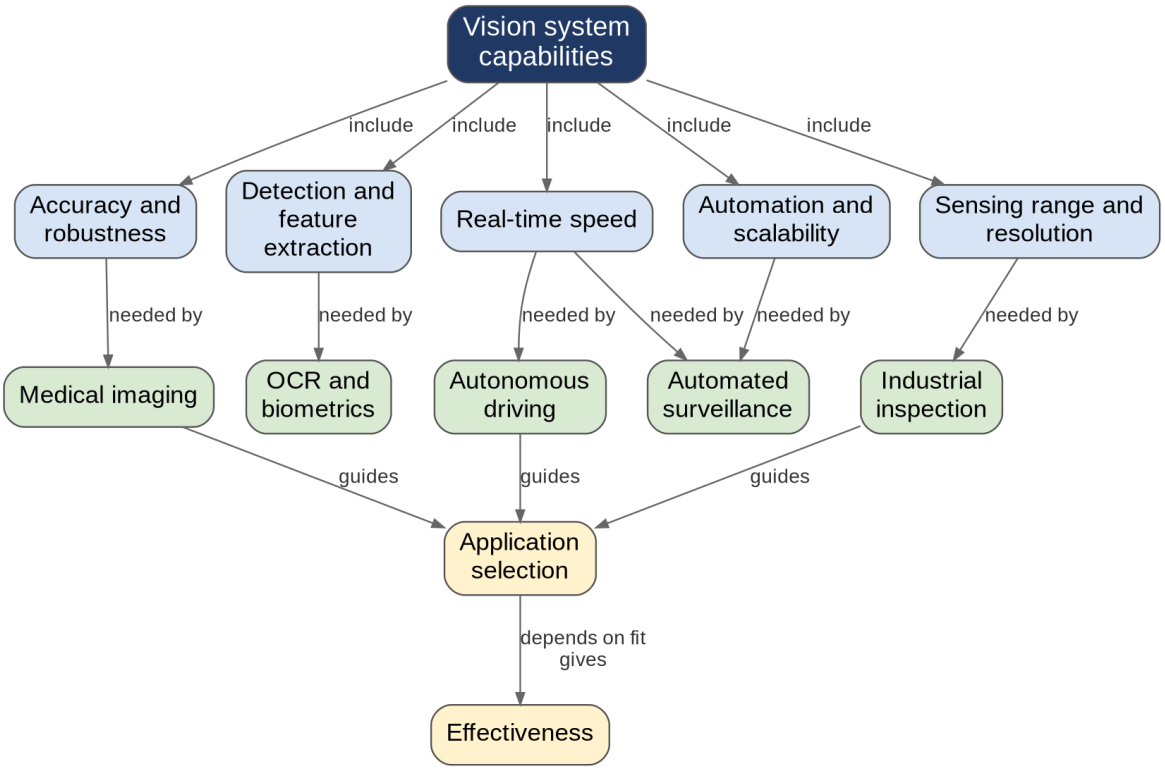
Literature link: *[1] defines spatial and intensity resolution (M×N pixels, L = 2^k gray levels) and shows false contouring and checkerboard effects when they are reduced.*

4. Acquisition conditions linked to downstream performance



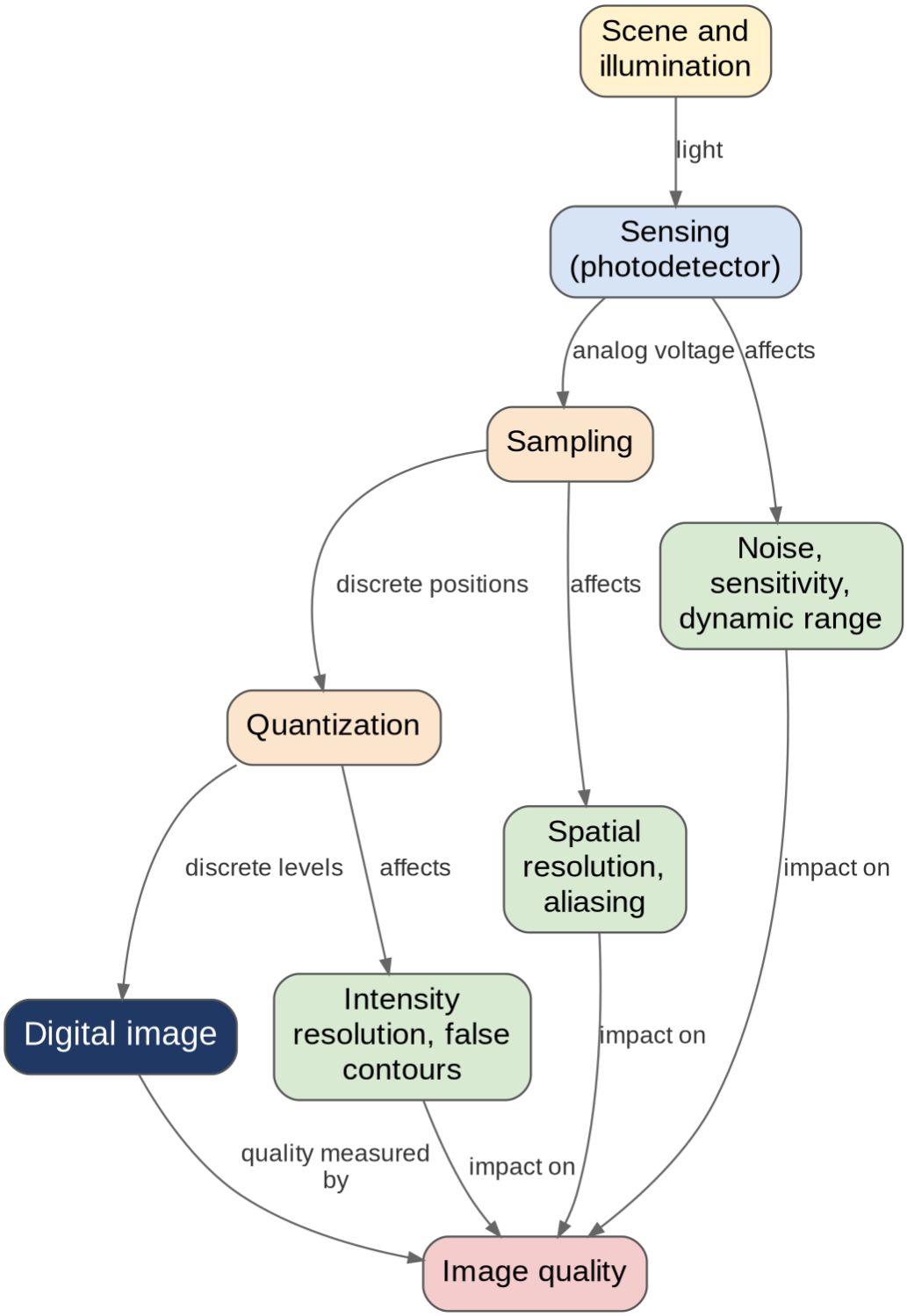
Literature link: *[2] discusses exposure, noise and motion blur arising at capture; [1] covers noise models and restoration; [7] covers sensor noise limits on usable signal.*

5. System capabilities and application effectiveness



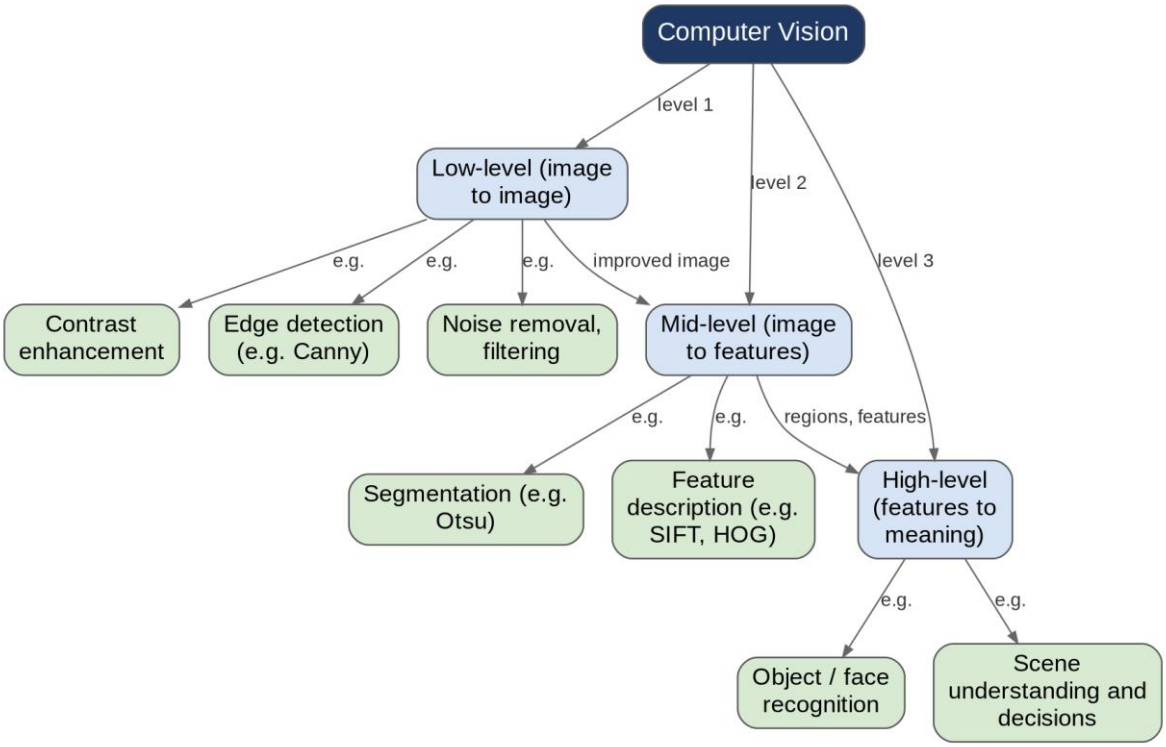
Literature link: *[3] surveys application areas (recognition, tracking, inspection, medical imaging); [2] discusses real-time speed and robustness requirements in practical systems.*

6. Workflow of digital image formation



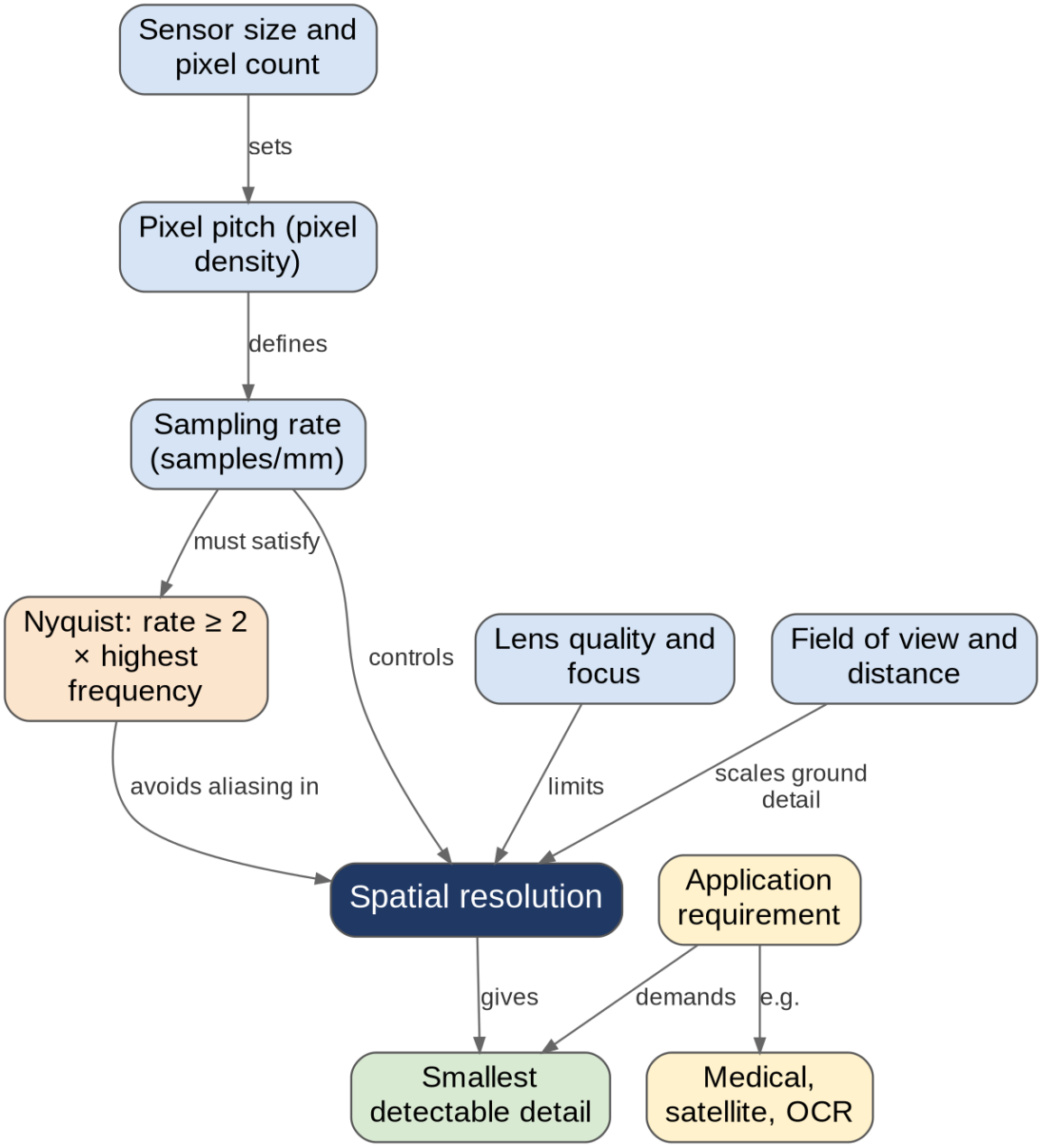
Literature link: *[1] and [8] describe sampling and quantization as the two steps of digitisation; [2] describes sensing and analog-to-digital conversion.*

7. Levels of Computer Vision



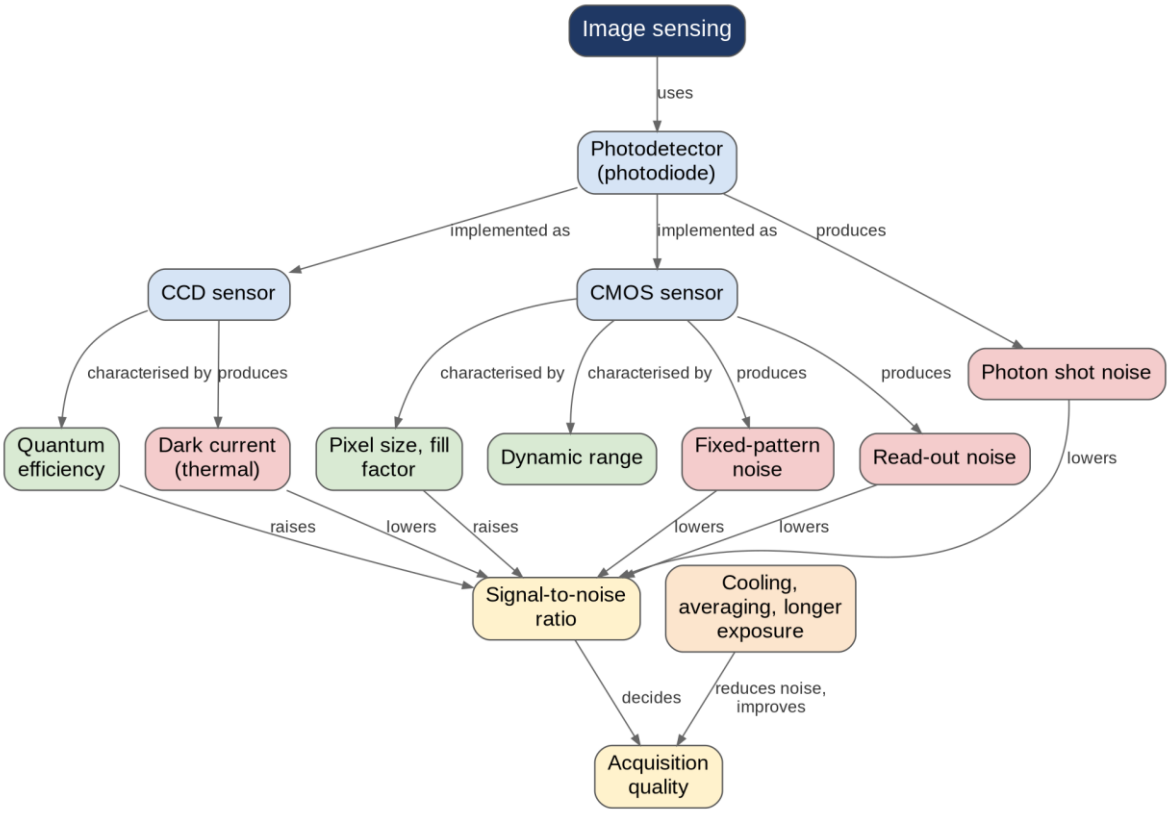
Literature link: *[4] separates early, intermediate and high-level vision; [1] describes low-, mid- and high-level processing as a continuum from image to understanding.*

8. Factors affecting spatial resolution



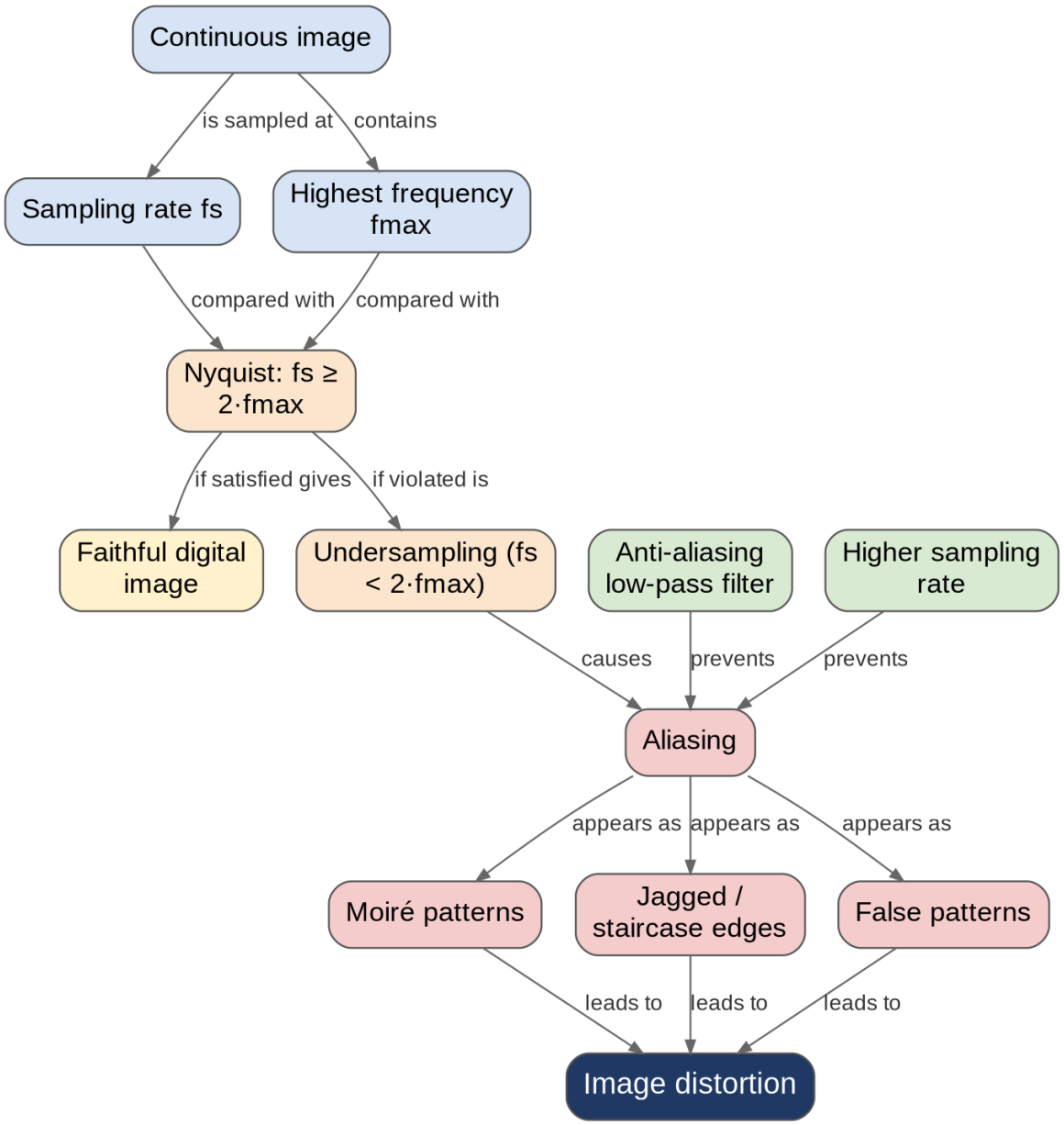
Literature link: *[1] relates spatial resolution to sampling and the Nyquist limit; [5] gives the sampling theorem; [2] covers lens blur and the modulation transfer function.*

9. Image sensing, sensor characteristics and noise



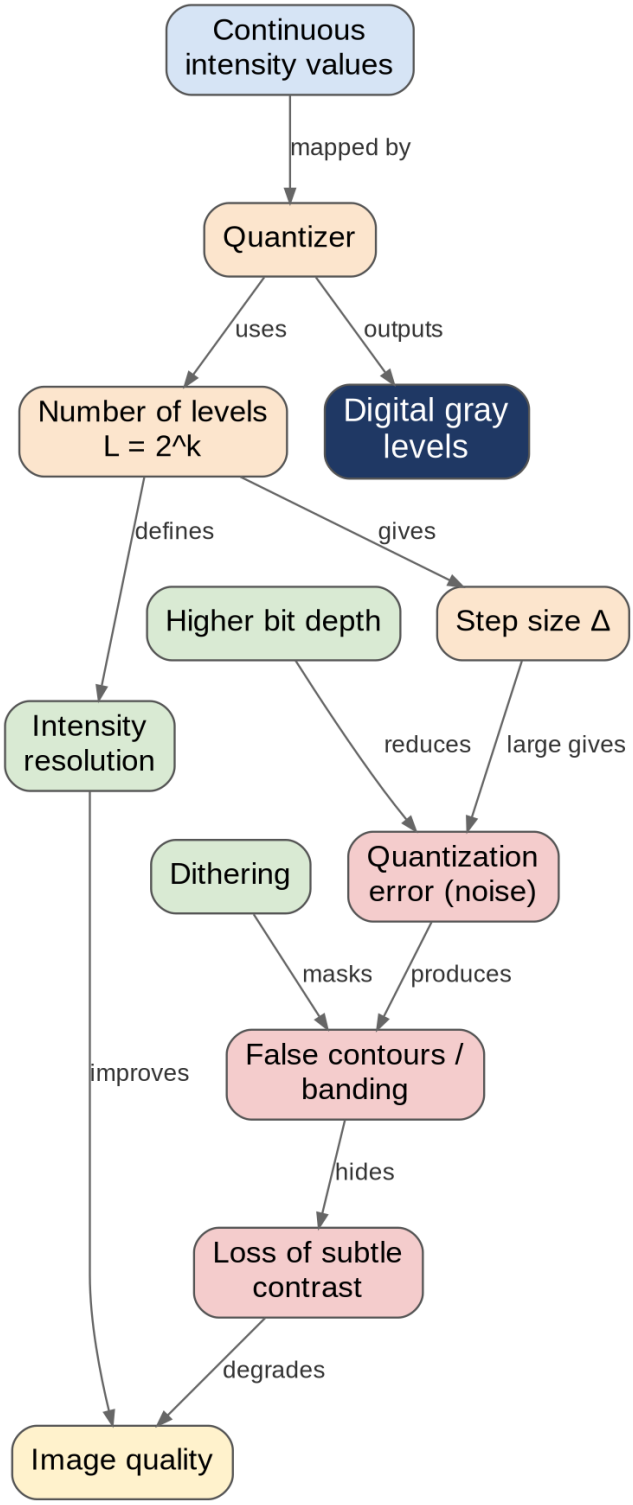
Literature link: *[6] compares CMOS and CCD architectures and their noise; [7] details CCD noise sources and signal-to-noise ratio; [2] gives a sensor noise model.*

10. Sampling, aliasing and image distortion



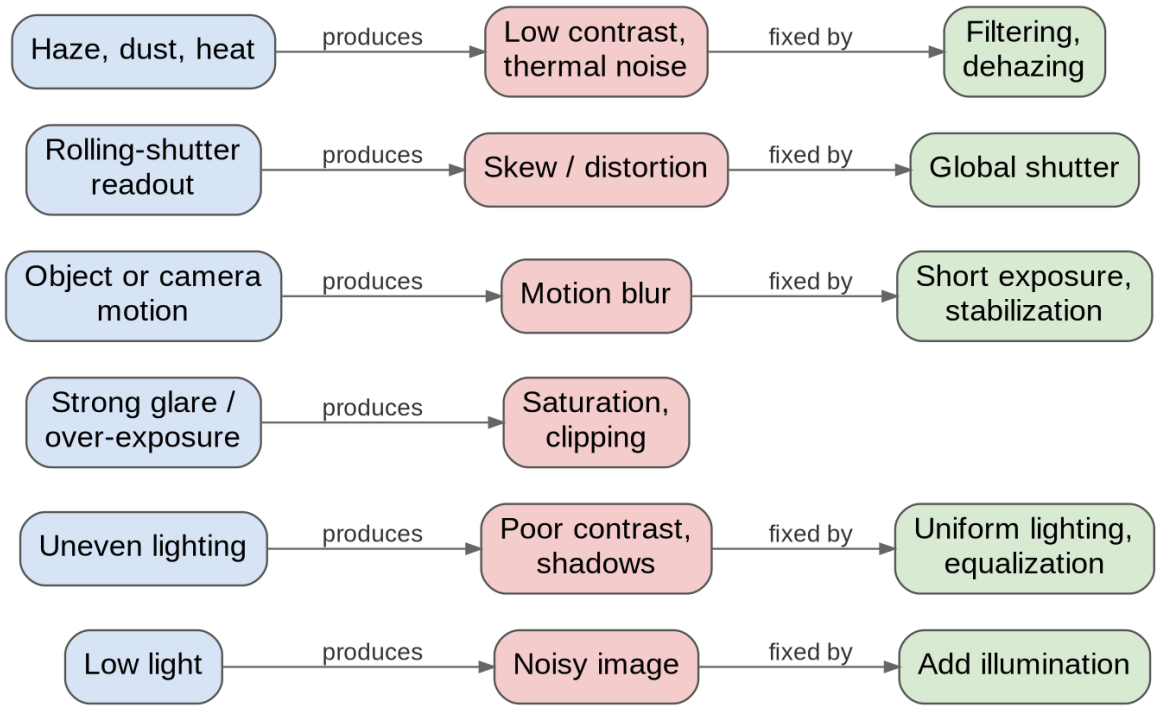
Literature link: *[5] gives the sampling theorem; [1] shows aliasing and moiré patterns in images and the use of low-pass filtering before resampling.*

11. Quantization and intensity resolution



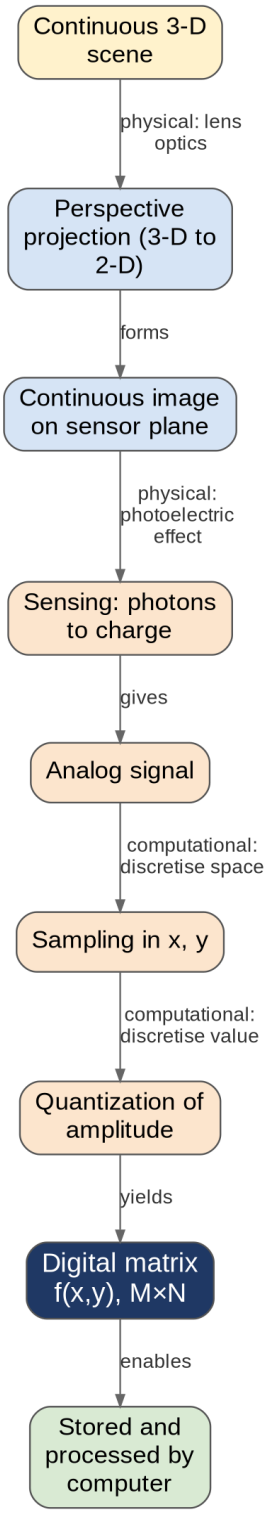
Literature link: *[1] explains uniform quantization, false contouring and the effect of gray-level count on perceived quality; [8] treats quantizer design and quantization error.*

12. Acquisition conditions and image artifacts



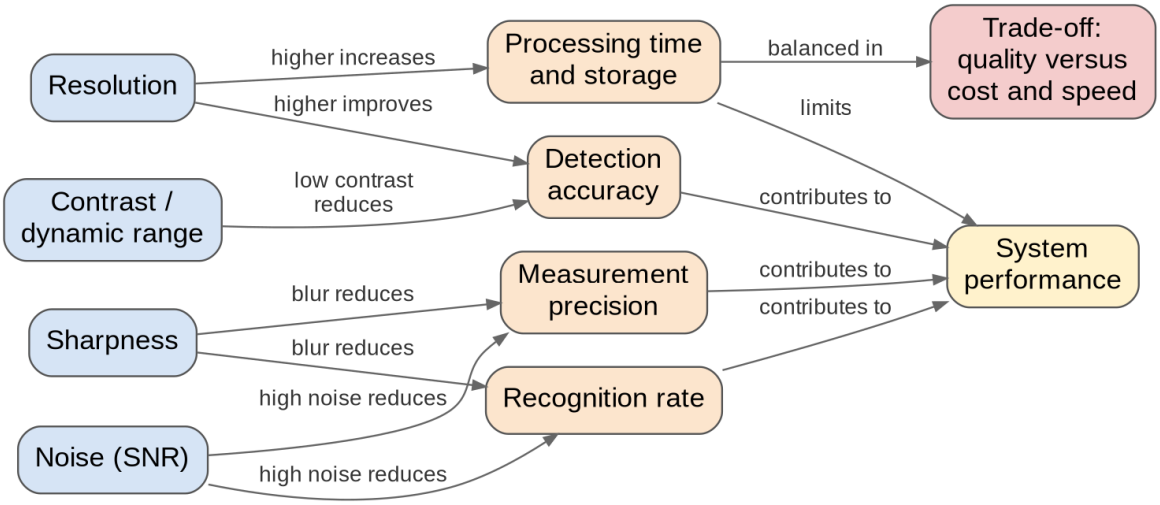
Literature link: *[2] covers exposure, motion blur, rolling shutter and noise; [3] covers illumination effects on image appearance.*

13. Transformation from analog scene to digital image



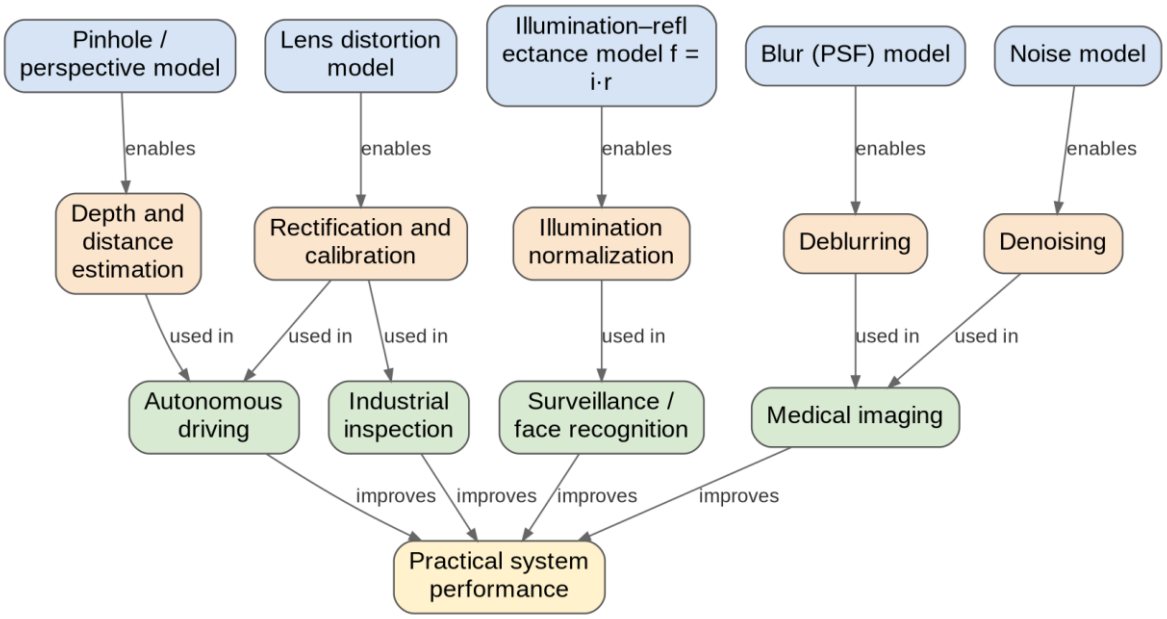
Literature link: *[1] derives the image model f(x,y) = i(x,y)·r(x,y) and the digitisation of continuous images; [2] covers perspective projection and sensing.*

14. Image quality parameters and system performance



Literature link: *[1] relates resolution, noise and contrast to enhancement and restoration; [3] discusses how image quality limits detection and recognition.*

15. Image formation models and Computer Vision applications



Literature link: *[2] and [3] treat pinhole projection, photometric models, lens distortion and calibration; [1] gives the illumination–reflectance and degradation models.*

References

[1] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018.

[2] R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed., Springer, 2022.

[3] D. A. Forsyth and J. Ponce, Computer Vision: A Modern Approach, 2nd ed., Pearson, 2012.

[4] D. Marr, Vision: A Computational Investigation into the Human Representation and Processing of Visual Information, W. H. Freeman, 1982.

[5] C. E. Shannon, “Communication in the presence of noise,” Proceedings of the IRE, vol. 37, no. 1, 1949.

[6] A. El Gamal and H. Eltoukhy, “CMOS image sensors,” IEEE Circuits and Devices Magazine, vol. 21, no. 3, 2005.

[7] J. R. Janesick, Scientific Charge-Coupled Devices, SPIE Press, 2001.

[8] A. K. Jain, Fundamentals of Digital Image Processing, Prentice Hall, 1989.